



Cyber Security of Power Grids

TU Delft EE4410

Dr. Alex Stefanov

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Syllabus: EE4410 Cyber Security of Power Grids

Course Manager / Lecturer

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Contact hours: Any request shall be done only via email to both responsible lecturer and instructors.

Kind request

- Please read carefully. This document overviews the organisational aspects of this course and provides answers to most of the questions you might have about the essence and structure of the course.
- The provided information might not be presented nor discussed in any of the lectures.

- The goal is that you are sufficiently informed prior to the first lecture, so you can appraise beforehand what to expect from every activity throughout the course. In case of doubt or additional questions, please do not hesitate to contact the responsible lecturer and instructors.

What do we expect from you?

- **Commitment**
 - Be prepared for the lectures. Study the discussed course material, i.e., lecture slides and references, and instructions for lab/practicum activities, before the lecture or lab/practicum session.
- **Eagerness to work independently**
 - Study all concepts and materials provided in the course and reference list.
- **Continuous communication**
 - Discussions outside the classroom and collaboration with peers (improvement of group performance) and your lecturer (questions, feedback, and improvement of the course).
- **Respect the intellectual property**
 - The material prepared and provided for this course is only for educational purposes.
 - It is discouraged to use it for other purposes without prior consent by the responsible lecturer.
 - **You are not allowed to reproduce nor distribute the course material without the written consent of the responsible lecturer!**

Disclaimer and ethics

- **All information provided in the course is for educational purposes only.** The lecturer(s) and instructor(s) are not responsible for any misuse of the information.
- The course does not promote and is strongly against illegal activities such as unethical hacking, cracking, and piracy.
- **Do not attempt to violate the law with anything contained in the course.** Do not abuse the material provided and be responsible.
- By participating in the course, **you (student) must accept that you will only lawfully use the material and knowledge you got from this course.**
- Any actions and / or activities related to the material contained within this course are solely your (student) responsibility. **The misuse of the information in this course can result in criminal charges brought against the persons in question.**

- The lecturer(s) and instructor(s) will not be held responsible in the event any criminal charges are brought against any individuals misusing the information from this course to break the law.

Course language: English

Expected prior knowledge: Basic knowledge of circuit theory and power system components is preferred although not necessary. **Optional:** SET3065 Intelligent Electrical Power Grids (Master EE / Master SET / Master EWEM)

Summary: The energy system transition towards a sustainable, low carbon future is led by decarbonisation, decentralisation, and digitalisation. Power grids are undergoing a fast-paced process of digitalisation for enhanced monitoring and control capabilities and grid intelligence. However, the increased integration of digital technologies, such as the next generation of Operational Technologies (OTs) and digital substations, implies a new risk as IT–OT systems for smart grids are vulnerable to cyber attacks. Examples of cyber security incidents related to power grids already exist around the world. On December 23, 2015 cyber attacks were conducted on the power grid in Ukraine that resulted in power outages, which affected 225,000 customers. More sophisticated cyber attacks on the Ukrainian power grid followed on December 17, 2016 resulting in a power outage in the distribution network where 200 MW of load was unsupplied. The complexity of cyber attacks on power systems is on the rise. They can initiate cascading failures and result in a catastrophic blackout. A power outage can disrupt the entire energy chain including water supply, heating, and gas networks. Without power, hospitals and other critical services are severely affected. A disruption of service may lead to financial loss, damages, chaos, or even a loss of lives. With respect to security of supply and reliability of the future energy system provision, special attention is needed for new vulnerabilities and cyber threats that come with digitalisation. Accordingly, the cyber security aspects are critical for a further power grid digitalisation.

This course aims to analyse and enhance the cyber security of smart grids. It analyses the typical IT–OT system architectures for power grid monitoring and control and explains the main cyber security concepts. The course provides essential knowledge to analyse cyber security threats, advanced persistent threats, threat vectors, and vulnerabilities of IT – OT systems for power grids. It examines the kill chain stages of the cyber attack process on power grids. The course analyses the impact of cyber attacks on power system operation in terms of loss of load, cascading effects, and risk of a blackout. The course studies the state-of-the-art security controls and best cyber security practices to defend against cyber attacks and effectively mitigate their impact on power system operation.

Course contents: This course focuses on cyber security of Information Technology (IT) – Operational Technologies (OT) networks used for power system operation. It starts with an overview of the smart grid and its industrial control systems (EMS/SCADA) and introduces computer networks and cyber security. Next, the course will introduce the cyber kill chain method for cyber security investigation and intelligence-driven defence. Kill chains are used to understand, anticipate, recognise, and combat cyber attacks and advanced persistent threats. The course uses the kill chains of the 2015/2016 cyber attacks on Ukrainian power grid as real-world study cases. The main IT-OT system vulnerabilities and cyber threats are analysed. The course covers threats such as passive and

active reconnaissance, social engineering, and malware. It investigates various threat vectors on host and network security and advanced cyber attacks targeting industrial control systems. The impact on power system operation is analysed. The course discusses the state-of-the-art security controls to defend and mitigate cyber attacks such as access controls, cryptography, firewalls, secure communication protocols, network segmentation, and intrusion detection and prevention systems. The best practices to improve power grid resilience to cyber attacks are discussed. This course provides a deep dive into core cyber security concepts such as cryptography (symmetric and asymmetric encryption), hash functions, digital signatures, and cyber-physical anomaly detection using machine learning techniques. Cyber security demonstrations will be conducted for students in Control Room of the Future (CRoF) technology centre.

Learning objectives

At the end of the course, you will be able to:

- LO1: Analyse cyber security threats, threat vectors, and vulnerabilities of IT–OT systems for power grids.
- LO2: Examine kill chain stages of the cyber attack process on cyber-physical power systems.
- LO3: Analyse the impact of cyber attacks on power system operation.
- LO4: Recommend appropriate security controls to detect, prevent and mitigate cyber attacks, and cyber secure power grids.

Education method

- 1) Lectures: theory and real-world study cases (8 lectures x 2h/lecture)
- 2) Practicums: hands-on computer labs (5 practicums x 2h/practicum)

Assignments

- All practicums have Assignments to be done in the lab and at home.
- For each practicum, students must complete an assignment. Each student must write an individual assignment report.
- The assignment reports (PDF) must be submitted on Brightspace (NO resubmission/iteration!), which will be checked automatically for plagiarism.
- The deadline for the report submission is one week after the practicum (lab session).
- All assignments and reports are mandatory.

- More information about the practicums, lab environment, assignments, and reports can be found in the Practicum Manual on Brightspace.

Time management

4 ECTS = 4 x 28h = 112h

- 16h (8 x 2h) lectures
- 10h (5 x 2h) practicums
- 86h self-study ($\approx 8 \times 9\text{h/week} + 14\text{h exam preparation}$)

Assessment

The grade is formed based on the following elements:

- 40% Assignment reports (five assignment reports submitted in time on Brightspace)
 - Figures/tables and text/analysis get a grade of 0 – 5 each (see Table 1) -> 0 – 10 points for each assignment.
 - Sum of all assignment points is divided by 5 -> 0 – 10 points in total.
 - Report submission not in time? -> 0 points.
- 60% Final digital (computer) exam on campus
 - Questions are of theoretical nature or simple calculations.
 - Multiple choice, open and/or closed form questions.
 - Approximately 25 questions.
 - 0 – 10 points in total.
- Maximum final grade is 10, you need ≥ 6 to pass the course.
- All assignments and their reports are mandatory.
- Grade rounded to .0 or .5 (e.g., 5.8 -> 6.0)

In case of an insufficient final result, repair options may exist in accordance with Article 2, Examination Requirements, Clause 4, of the Implementation Regulations 2023-2024.

Disclaimer: information may change depending on unforeseen circumstances or measures (see: TER Art 29, sub 4).

Table 1. Assignment Assessment Matrix

Grade	Quality of figures and tables	Quality of text and analysis
0	IF IN ALL FIGURES/TABLES: Labels/captions/legend missing. Illegible text. Overcrowding. Unidentifiable elements (e.g., curves). Poor resolution.	Incomplete description of the concept and results. No reflections. Document contains serious spelling and grammatical errors. Report with more than one page over-length.
1	IF IN AT LEAST TWO FIGURES/TABLES: Labels/captions/legend missing. Illegible text. Overcrowding. Unidentifiable elements (e.g., curves). Poor resolution.	Follows the indicated structure, fair descriptions of the concept and results. No reflections. Document contains some spelling and grammatical errors. Report with more than one page over-length.
2	IF IN AT LEAST ONE FIGURE/TABLE: Labels/captions/legend missing. Illegible text. Overcrowding. Unidentifiable elements (e.g., curves). Poor resolution.	Follows the indicated structure, proper description of concepts and results. Poor reflections on results. The document contains few spelling and grammatical errors. Report with one page over-length.
3	Good quality figures/tables. However, information is not properly processed. Figures and tables have unnecessary info.	Follows the indicated structure, good description of concepts and results. Limited reflections on results. Document contains only minor spelling and grammatical errors. Report with one page over-length.
4	Very good quality figures/tables. However, information is not properly processed. Figures and tables contain essential info.	Very good in terms of structure, contents, clarity, analysis, and reflections. Concepts from the lecture are understood and used to analyse the results. Document is without any spelling and grammatical errors. Report with the correct length.
5	Excellent quality figures/tables. Excellent processing and presentation of results. Only essential information is provided to convey the main findings properly.	Excellent in terms of structure, contents, clarity, analysis, and reflections. Concepts from the lecture are understood and used to

		analyse the results. Spelling and grammatically error-free. Report with the correct length.
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Sample questions for the exam

Table 2 provides examples of exam questions.

Table 2. Examples of exam questions

1	What are Operational Technologies (OT)?	IT systems for power grids Information and communication technologies used for energy markets Information and communication technologies used for power grid monitoring and control
2	Substations communicate with the power system control centre via	Public communication networks The Internet Private communication networks
3	The wide area communication network is considered to be a	Trusted network Semi-trusted network Untrusted network
4	What security control is typically used to segregate the IT and OT systems at control centres for power grids?	Firewall Encryption Intrusion detection system
5	Cyber attacks on power grids are	Low impact, low frequency disturbances High impact, high frequency disturbances High impact, low frequency disturbances

6	What kind of facilities did Stuxnet malware target?	Nuclear Hydro power plants Windfarms
7	What was the impact of the Ukraine 2015 cyber attack?	Power outages affecting 225,000 customers Complete blackout affecting 500,000 customers Power system instability
8	How did attackers intrude in the utility IT system in the Ukraine 2015 cyber attack?	By using phishing e-mails By stealing credentials By using an infected USB
9	What is the impact of injecting spoofed IEC 61850 GOOSE traffic into the substation communication network for power grids?	Maliciously modify power generation Maliciously open circuit breakers to disconnect transmission lines Maliciously change voltage setpoints
10	What is the recommendation of IEC 62351-6 standard to secure the IEC 61850 protocols?	Use HMAC to check the integrity of the IEC 61850 data frames Use virtual private networks Use firewalls and antivirus in substations
11	The Internet backbone consists of a group of regional ISPs.	True False
12	Which of the following elements are placed at critical points within a system to protect vital assets?	Firewall IDS IPS All of the above
13	Identify all deployment requirements that should be considered when determining the locations and features of firewalls.	Write your answer here.

14	Describe the limitations of firewall inspection.	Write your answer here.
15	Describe and compare the key functions of IDS and IPS.	Write your answer here.

Required reading material for lectures and practicums

- This information is subject to changes. Changes will be announced in advance on Brightspace.
- **Lecture slides used during the lectures are always included in the required reading material.**
- The books, papers, articles and/or reports referred to here are additional to lecture slides. When page numbers are specified, you only need to read the specified pages. When no page numbers are specified, the reading refers to the entire section, chapter, or publication.
- No need to buy the books or the other reading materials. You can read them in the library. They are also free online via the TU Delft Library when accessed from within the TU Delft network.
- The books can be accessed online (free) on the TU Delft Library using the links provided in the Literature section below.

Lectures and practicums

April 22, 2025 15:45 – 17:45 Lecture 1: Introduction to Computer Networks		
#	Topic	Suggested reading material
1	Smart grids and cyber security	Knapp, E. Applied cyber security and the smart grid (Chapter 2)
2	Introduction to computer networks and The Internet architecture	Wu, C. Introduction to computer networks and cybersecurity (Chapter I.2, I.3, I.4)
3	Packet switching, packet loss, delays, and congestion	Wu, C. Introduction to computer networks and cybersecurity (Chapter I.5, I.6)

4	The protocol stack	Wu, C. Introduction to computer networks and cybersecurity (Chapter I.7, I.8)
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April 29, 2025 15:45 – 17:45 Lecture 2: IT-OT Communication Networks for Power Grids and Cyber Security		
#	Topic	Suggested reading material
1	IT-OT communication networks for power grids	Knapp, E. Applied cyber security and the smart grid (Chapter 2)
2	SCADA vulnerabilities, threats and attack vectors	Knapp, E. Industrial network security (Chapter 2, 3, 6); ENISA, Communication network dependencies for ICS/SCADA systems, 2017; Knapp, E. Applied cyber security and the smart grid (Chapter 2)
3	Introduction to cyber security	Wu, C. Introduction to computer networks and cybersecurity (Chapter I.9.2, Chapter 17.1, 17.2)
4	Cyber kill chain and security controls	Wu, C. Introduction to computer networks and cybersecurity (Chapter I.9.2, Chapter 17.1, 17.2)

May 6, 2025 15:45 – 17:45 Lecture 3: Cryptography		
#	Topic	Suggested reading material
1	Symmetric encryption	N.P. Smart. Cryptography Made Simple (Chapter 7.1 [pgs. 119-120, 121 - 132], 10.1 [pgs. 179-182], 12.4 [pgs. 238, 239], 13.1 to 13.5 [pgs. 241, 244, 245, 250, 252, 253, 254, 255, 266])
2	Message authentication codes, hashing	N.P. Smart. Cryptography Made Simple (Chapter 11.7.1, Chapter 14.1 [pgs. 271, (273,274: Cryptographic Hash function properties)], 14.5 [pgs. 282, 283])
3	Asymmetric encryption	N.P. Smart. Cryptography Made Simple (Chapter 11.4 [pg. 202], 15.1[pgs. 295, 296], 16.2.1 [pg. 321])

4	Digital signatures	N.P. Smart. Cryptography Made Simple (Chapter 11.7.2 [pg. 216])
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May 13, 2025 15:45 – 17:45 Lecture 4: Cyber Attacks on IT-OT Communication Networks		
#	Topic	Suggested reading material
1	Cyber attacks on IT systems	Baloch, R. Ethical hacking and penetration testing guide (Chapter 3); Dunkerley, D. CompTIA security+ exam sy0-501 (Chapter 2.02); Wu, C. Introduction to computer networks and cybersecurity (Chapter 17.4); Baloch, R. Ethical hacking and penetration testing guide (Chapters 3, 4, 5, 6); Dunkerley, D. CompTIA security+ exam sy0-501 (Chapter 9.03)
2	Cyber attacks on IEC 61850 in digital substations	Papers: IEC 61850 attacks
3	Cyber-physical system data correlation	Abedi, V. S. Rajkumar, A. Ştefanov and P. Palensky, "Towards Real-Time Distinction of Power System Faults and Cyber Attacks"
4	Study case: cyber attacks on power grid in Ukraine 2015	Lee, R. Analysis of the cyber attack on the Ukrainian power grid

May 14, 2025 10:45 – 12:45 Lab Session 1: URL Phishing, Malware, and Passive Reconnaissance		
#	Topic	Suggested reading material
1	Objectives	Practicum manual
2	URL phishing and malware identification	Practicum manual; Dunkerley, D. CompTIA security+ exam sy0-501 (Chapter 2.02, 11.01); Wu, C. Introduction to computer networks and cybersecurity (Chapter 17.3, 17.4, 20.2, 26.4);
3	Passive reconnaissance	Practicum manual; Dunkerley, D. CompTIA security+ exam sy0-501 (Chapter 2.02, 11.01); Maltego User Guide
4	Assignment 1	Practicum manual

May 20, 2025 15:45 – 17:45 Lecture 5: Impact Analysis of Cyber Attacks on Power System Operation: Cascading Failures and Blackout		
#	Topic	Suggested reading material
1	Power system dynamics: modelling and simulation	Power Systems Stability and Control 2 nd Edition by Prabha S. Kundur, 2022. Part 3, Transient stability.
2	Cascading failures and blackouts	V. S. Rajkumar et al., "Cyber Attacks on Power Grids: Causes and Propagation of Cascading Failures," doi:10.1109/ACCESS.2023.3317695. P. Pourbeik et al., "The anatomy of a power grid blackout - Root causes and dynamics of recent major blackouts," doi: 10.1109/MPAE.2006.1687814. G. Andersson et al., "Causes of the 2003 major grid blackouts in North America and Europe, and recommended means to improve system dynamic performance," doi: 10.1109/TPWRS.2005.857942.
3	Cyber attack impact analysis	Papers: IEC 61850 attacks V. S. Rajkumar et al., "Cyber Attacks on Power Grids: Causes and Propagation of Cascading Failures," doi:10.1109/ACCESS.2023.3317695.

May 21, 2025 10:45 – 12:45 Lab Session 2: Cryptography Applications		
#	Topic	Suggested reading material
1	Objectives	Practicum manual; Jupyter notebook
2	Symmetric encryption	Practicum manual; Jupyter notebook; Lecture 3 slides; 13.1 to 13.5 [pgs. 241, 244, 245, 250, 252, 253, 254, 255, 266]); PyCryptodome Lib docs

3	Asymmetric encryption	Practicum manual; Jupyter notebook; Lecture 3 slides; N.P. Smart. Cryptography Made Simple (Chapter 11.4 [pg. 202], 15.1[pgs. 295, 296], 16.2.1 [pg. 321]); PyCryptodome Lib docs
4	Digital signatures	Practicum manual; Jupyter notebook; Lecture 3 slides; N.P. Smart. Cryptography Made Simple (Chapter 11.7.2 [pg. 216]); PyCryptodome Lib docs
5	Assignment 2	Practicum manual; Jupyter notebook

May 27, 2025 15:45 – 17:45 Lecture 6: Dynamical Analysis of Cyber Induced Power System Cascading Failures		
#	Topic	Suggested reading material
1	Dynamical analysis of cascading failures	V.S. Rajkumar, "Dynamical analysis of power system cascading failures caused by cyber attacks" T. J. Browne et.al., "Practical Application of Hilbert Transform Techniques in Identifying Inter-area Oscillations," in Inter-area Oscillations in Power Systems, Springer. Doi: 10.1007/978-0-387-89530-7_4
2	Point of no return identification and quantification	V.S. Rajkumar, "Dynamical analysis of power system cascading failures caused by cyber attacks" P. Pourbeik et al., "The anatomy of a power grid blackout - Root causes and dynamics of recent major blackouts," doi: 10.1109/MPAE.2006.1687814.
3	Experimental results	V.S. Rajkumar, "Dynamical analysis of power system cascading failures caused by cyber attacks"

May 28, 2024 10:45 – 12:45 Lab Session 3: Cyber-Physical Data Correlation		
#	Topic	Suggested reading material

1	Objectives	Practicum manual
2	Anomaly detection and correlation of power grid cyber-physical data	Practicum manual; Abedi, V. S. Rajkumar, A. Ştefanov and P. Palensky, "Towards Real-Time Distinction of Power System Faults and Cyber Attacks"
3	Assignment 3	Practicum manual

June 3, 2025 15:45 – 17:45 Lecture 7: Anomaly Detection and Cyber Attacks Mitigation		
#	Topic	Suggested reading material
1	IT-OT network segmentation, intrusion detection and prevention systems	
2	Advanced persistent threats on power grids	A. Presekal, "Attack Graph Model for Cyber-Physical Power Systems Using Hybrid Deep Learning"; A. Presekal, "Cyber Forensic Analysis for Operational Technology Using Graph-Based Deep Learning"
3	Cyber-physical system model	A. Presekal, "Attack Graph Model for Cyber-Physical Power Systems Using Hybrid Deep Learning"; A. Presekal, "Cyber Forensic Analysis for Operational Technology Using Graph-Based Deep Learning"
4	Traffic dispersion graphs	- Presekal, "Attack Graph Model for Cyber-Physical Power Systems Using Hybrid Deep Learning" A. Presekal, "Cyber Forensic Analysis for Operational Technology Using Graph-Based Deep Learning" - Iliofotou, Marios, et al. "Network monitoring using traffic dispersion graphs (tdgs)." Proceedings of the 7th ACM SIGCOMM conference on Internet measurement. 2007

June 4, 2025 10:45 – 12:45 Lab Session 4: Operational Technology Traffic Analysis		
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#	Topic	Suggested reading material
1	Objectives	Practicum manual
2	OT networks	Practicum manual; A. Presekal, "Attack Graph Model for Cyber-Physical Power Systems Using Hybrid Deep Learning"; A. Presekal, "Cyber Forensic Analysis for Operational Technology Using Graph-Based Deep Learning"
3	Traffic dispersion graphs	Practicum manual; A. Presekal, "Attack Graph Model for Cyber-Physical Power Systems Using Hybrid Deep Learning"; A. Presekal, "Cyber Forensic Analysis for Operational Technology Using Graph-Based Deep Learning"
4	Assignment 4	Practicum manual

June 10, 2025 15:45 – 17:45 Lecture 8: Anomaly Detection and Cyber Attacks Mitigation		
#	Topic	Suggested reading material
1	Graph convolutional long short-term memory	A. Presekal, "Attack Graph Model for Cyber-Physical Power Systems Using Hybrid Deep Learning"; A. Presekal, "Cyber Forensic Analysis for Operational Technology Using Graph-Based Deep Learning"
2	Time series classification with convolutional neural network	A. Presekal, "Attack Graph Model for Cyber-Physical Power Systems Using Hybrid Deep Learning"; A. Presekal, "Cyber Forensic Analysis for Operational Technology Using Graph-Based Deep Learning"
3	Anomaly detection strategies	A. Presekal, "Attack Graph Model for Cyber-Physical Power Systems Using Hybrid Deep Learning"; A. Presekal, "Cyber Forensic Analysis for Operational Technology Using Graph-Based Deep Learning"
4	Attack graph model	A. Presekal, "Attack Graph Model for Cyber-Physical Power Systems Using Hybrid Deep Learning"; A. Presekal, "Cyber

		Forensic Analysis for Operational Technology Using Graph-Based Deep Learning"
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June 11, 2025 10:45 – 12:45 Lab Session 5: Anomaly Detection and Time Series Classification		
#	Topic	Suggested reading material
1	Objectives	Practicum manual
2	Cyber-physical power system co-simulation	Practicum manual; A. Presekal, "Attack Graph Model for Cyber-Physical Power Systems Using Hybrid Deep Learning"; A. Presekal, "Cyber Forensic Analysis for Operational Technology Using Graph-Based Deep Learning"
3	Anomaly detection and time series classification	Practicum manual; A. Presekal, "Attack Graph Model for Cyber-Physical Power Systems Using Hybrid Deep Learning"; A. Presekal, "Cyber Forensic Analysis for Operational Technology Using Graph-Based Deep Learning"
4	Assignment 5	Practicum manual

June 23, 2025 09:00 – 12:00 Digital Exam		
#	Digital (computer) exam: with registration on campus	

July 14, 2025 09:00 – 12:00 Resit Digital Exam		
#	Digital (computer) exam: with registration on campus	

Literature

Books: cyber security & industrial network security

1. Wu, Chwan-Hwa, & Irwin, J. David. (2013). **Introduction to computer networks and cybersecurity**. CRC Press. ISBN: 9781498760133. Link: <https://tudelft.on.worldcat.org/oclc/899501242>
 - a. This book can be accessed online (free) on the TU Delft Library using the provided link above.
 - b. This book can also be purchased at a discounted price for TU Delft students on Studystore.nlLink: <https://www.studystore.nl/p/9781466572133/introduction-to-computer-networks-and-cybersecurity>
2. Nigel P. Smart. (2016). **Cryptography made simple**. Springer, 2nd Edition. Link: <https://link-springer-com.tudelft.idm.oclc.org/book/10.1007/978-3-319-21936-3>
3. Dunkerley, Dawn, & Meyers, Mike. (2018). **CompTIA security+ exam sy0-501 (Fifth Edition, Ser. Mike Meyers' certification passport)**. Mc Graw Hill Education. ISBN: 9781260026559. Link: <https://tudelft.on.worldcat.org/oclc/1015247902>
4. Bishop, Matt, Sullivan, Elisabeth, & Ruppel, Michelle. (2019). **Computer security: art and science (Second Edition)**. Addison-Wesley. ISBN: 9780134097145. Link: <https://tudelft.on.worldcat.org/oclc/1082143685>
5. Knapp, Eric D., & Langill, Joel Thomas. (2015). **Industrial network security: securing critical infrastructure networks for smart grid, SCADA, and other industrial control systems (Second Edition)**. Syngress. ISBN 9780124201149. Link: <https://tudelft.on.worldcat.org/oclc/898069530> (ScienceDirect: <https://www-sciencedirect-com.tudelft.idm.oclc.org/book/9780124201149/industrial-network-security>)
6. Knapp, Eric D., Samani, Raj, & Langill, Joel. (2013). **Applied cyber security and the smart grid: implementing security controls into the modern power infrastructure**. Syngress. ISBN 978-1-59749-998-9. Link: <https://tudelft.on.worldcat.org/oclc/850602985> (ScienceDirect: <https://www-sciencedirect-com.tudelft.idm.oclc.org/book/9781597499989/applied-cyber-security-and-the-smart-grid>)
7. Baloch, Rafay. (2015). **Ethical hacking and penetration testing guide**. CRC Press. ISBN: 9781315145891. Link: <https://tudelft.on.worldcat.org/oclc/1011595255>

Papers:

1. Subramaniam Rajkumar, V., Stefanov, A., Rueda, J. L., & Palensky, P. (2024). Dynamical Analysis of Power System Cascading Failures Caused by Cyber Attacks. *IEEE Transactions on Industrial Informatics*, 1-11. <https://doi.org/10.1109/TII.2024.3372024> (open access).
2. Subramaniam Rajkumar, V., Stefanov, A., Presekal, A., Palensky, P., & Rueda, J. L. (2023). Cyber Attacks on Power Grids: Causes and Propagation of Cascading Failures. *IEEE Access*, 11, 103154-103176. <https://doi.org/10.1109/ACCESS.2023.3317695> (open access).
3. Presekal, A., Stefanov, A., Subramaniam Rajkumar, V., & Palensky, P. (2023). Attack Graph Model for Cyber-Physical Power Systems Using Hybrid Deep Learning. *IEEE Transactions on Smart Grid*, 14(5), 4007-4020. <https://doi.org/10.1109/TSG.2023.3237011>. Link: <https://research.tudelft.nl/en/publications/attack-graph-model-for-cyber-physical-power-systems-using-hybrid->
4. Subramaniam Rajkumar, V., Tealane, M., Stefanov, A., Presekal, A., & Palensky, P. (2020). **Cyber attacks on power system automation and protection and impact analysis**. Proc. IEEE PES ISGT, Delft, Netherlands, October 2020. Link: <http://resolver.tudelft.nl/uuid:660b7aa5-3526-4396-a6ca-d40451f4743f>
5. Subramaniam Rajkumar, V., Tealane, M., Stefanov, A., & Palensky, P. (2020). **Cyber attacks on protective relays in digital substations and impact analysis**. Proc. IEEE PES MSCPES, Virtual Workshop, April 2020. Link: <http://resolver.tudelft.nl/uuid:9c42eb78-da12-4818-bda4-5ec390cd562a>
6. P. Pourbeik, P. S. Kundur and C. W. Taylor, "The anatomy of a power grid blackout - Root causes and dynamics of recent major blackouts," in *IEEE Power and Energy Magazine*, vol. 4, no. 5, pp. 22-29, 2006, doi: 10.1109/MPAE.2006.1687814.

Reports / White Papers: Ukraine 2015 & 2016

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